

ABSTRACT

AI-POWERED SUPPLY CHAIN CONVERSATIONAL KNOWLEDGE ASSISTANT FOR SMALL AND MEDIUM ENTERPRISES

Small and Medium Enterprises (SMEs) often face difficulties in managing and accessing critical supply chain information due to the absence of advanced information systems and dedicated expertise. Important organizational knowledge such as procurement procedures, inventory policies, supplier information, warehouse guidelines, logistics records, and standard operating procedures is frequently distributed across multiple documents, making information retrieval inefficient and time-consuming.

This project proposes an AI-powered conversational knowledge assistant that enables employees and managers to access company knowledge through natural language interaction. The system utilizes Retrieval-Augmented Generation (RAG) technology to retrieve relevant information from verified organizational documents before generating responses, ensuring accurate answers with source references. In addition, an integrated supply chain dashboard provides operational insights including inventory status, low-stock alerts, supplier details, purchase order status, stock movement trends, and warehouse activities.

The proposed solution offers SMEs an affordable and user-friendly alternative to expensive enterprise systems while improving knowledge accessibility, employee onboarding, knowledge preservation, and operational efficiency through a single intelligent platform.

Keywords: *Retrieval-Augmented Generation (RAG), Supply Chain Management, Conversational AI, Knowledge Management, Small and Medium Enterprises (SMEs).*

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